

Smart Seaweed Cultivation Assistant – Seaweed Type Identification and Classification

25-26J-311

Project Proposal Report

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B.Sc. (Hons) in Information Technology Specialized in Information Technology

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Declaration Page of the Candidates & Supervisor

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Date

Abstract

Manual identification of seaweed species in Sri Lanka is often challenging for amateur farmers and coastal communities, which leads to mistakes such as harvesting the wrong species or harvesting non-edible and low-value species [1]. Such misidentification not only reduces income but can also impact food safety negatively. To overcome this shortcoming, this research proposes the development of an AI-based, image-classification system to identify two valuable Sri Lankan seaweed species: *Kappaphycus alvarezii* and *Gracilaria edulis*. These seaweeds are commercially valuable, widely cultivated in local waters, and in great demand in food and industrial applications [2], [3].

The platform will allow users to upload photos of seaweed captured in natural coastal environments or after harvesting. The photos will be pre-processed using computer vision techniques and then run through a Convolutional Neural Network (CNN) [7] model that has been trained on a carefully curated dataset of seaweed photos. A "non-seaweed" class will also be included to reject undesirable inputs so that farmers and communities receive accurate, trustworthy results [8]. A structured knowledge base will contain useful information about each species, including growth habits, edibility, nutritional value [3], and economic value, and will be accessible through a user-friendly web interface.

Aside from the technological innovation, this research has two broad social objectives. First, it strengthens vegetarian communities of people by promoting seaweed as a protein-, mineral-rich, and sustainable food option [3]. Second, it aims to empower low-income households, fishing villages, coastal women, and especially those with no stable income by giving them a low-cost and easy-to-use tool to improve seaweed cultivation. Through enabling identification and supporting culturing, the project supports livelihood development, food security, and the growth of Sri Lanka's blue economy [2], [6].

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Table 1: List of Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
CNN	Convolutional Neural Network
ML	Machine Learning
DB	Database
GPU	Graphics Processing Unit
UI	User Interface
SLIIT	Sri Lanka Institute of Information Technology

1. Introduction

1.1 Background & Literature Survey

Seaweed cultivation is increasingly viewed as a sustainable and economically valuable sector in Sri Lanka's blue economy. Due to more than 800 km of coastline and the changing marine ecosystems, Sri Lanka has a wide range of macroalgae species, some of which are of enormous commercial value in the global market. Seaweeds are used widely in the food industry as a source of protein diet, in pharmaceuticals and cosmetics because of their bioactive principles, and as raw materials to produce hydrocolloids such as agar and carrageenan [1], [3]. Besides their economic importance, seaweeds are also of outstanding environmental significance with functions such as carbon sequestration, nutrient cycling, and coastal ecosystem protection.

Protein scarcity remains a major nutrition issue in Sri Lanka, particularly among poor households that lack adequate low-cost sources of protein. Seaweeds such as *Kappaphycus alvarezii* and *Gracilaria edulis* are rich in nutrients and contain high protein, minerals, and bioactive substances, which can alleviate this shortage [3]. Through seaweed production and utilization, Sri Lanka can improve food security and help vegetarian and coastal communities with sustainable, low-cost protein options.

From the researcher's perspective, existing research indicates Sri Lankan seaweed's economic and environmental significance [2], [3] and new AI models such as CNNs [7], [10] and Mask R-CNN [4] have shown capability to excel in plant and ocean classification tasks. This research lays the foundation for AI-driven classification as feasible and highlights the non-availability of localized farmer-centric systems.

1.2 Research Gap

Despite all the materials available and existing research, there are gigantic loopholes:

Existing recognition apps are not specific to Sri Lankan seaweed species [6].

No integration of AI classification with localized knowledge bases for edibility, nutritional content, and farming practices.

No system to confidently reject non-seaweed inputs (false positives).

No end-to-end farmer-friendly system combining CNN classification with a full-stack web application.

These gaps highlight the need for a locally embraced, AI-driven identification tool for Sri Lankan seaweed farmers.

Study/Source	Focus	Limitation	Gap for Current Study
Coppejans et al. (2009) [1]	Taxonomy of Sri Lankan seaweeds	Requires expert morphological skills	No digital/automated tool for farmers
Premarathna et al. (2020) [2]	Distribution & ecology of seaweeds	Focused on ecology, not classification	No user-friendly identification system
Rajapaksha et al. (2023) [3]	Nutritional value of seaweeds	Nutritional study only	No AI-based classification link
Chatterjee et al. (2023) [4]	Mask R-CNN for corals	Limited to corals, not seaweeds	Need local dataset + CNN for seaweed
Lee & Tan (2024) [5]	MobileNetV3 for kelp/seagrass	No Sri Lankan seaweed dataset	Transfer learning required
Current Study	CNN-based classification	—	First localized AI system for Kappaphycus & Gracilaria

Figure 1 Research Gap Analysis of Existing vs Proposed System

1.3 Research Problem

The principal research problem is the absence of an easy-to-use, precise, and real-time digital identifier for Sri Lankan seaweed species. Farmers now use manual observation, morphological taxonomy, or oral information, which are prone to error, unreliable, and cause misclassification. It results in decreased cultivation efficiency, economic losses, and even potential health issues in case of harvesting non-edible species.

1.4 The main research question

How to create an image recognition system using AI to properly identify and classify Sri Lankan seaweed species (*Kappaphycus alvarezii* and *Gracilaria edulis*) and reject non-seaweed inputs, and integrate this with a localized farmers' knowledge base?

2. OBJECTIVES

2.1 Main Objective

The main purpose of this research is to conceptualize and develop an AI-activated seaweed identification and classification system using machine learning and computer vision techniques to provide immediate, accurate classification of Sri Lankan seaweeds. The system will be integrated with a question-based knowledge database to facilitate easy retrieval of important information on edibility, nutritional value [3], uses, and cultivation.

2.2 Specific Objectives

For the achievement of the ultimate objective, the following specific objectives are defined:

Dataset Development – Collect, label, and condition a dataset for seaweed species from the local area, such as *Kappaphycus alvarezii* and *Gracilaria edulis*, and non-seaweed material (e.g., leaves, rocks, trash) to enable "false input" rejection.

Model Training and Evaluation – Train and validate a Convolutional Neural Network (CNN) [7] model in Python using TensorFlow/PyTorch to predict seaweed species with maximum accuracy, supported by performance metrics such as accuracy, precision, recall, and F1-score.

Image Preprocessing – Apply OpenCV preprocessing techniques (resizing, filtering, noise removal, and contour sharpening) to improve image quality and ensure model stability against real-world conditions (e.g., underwater blur, varying outdoor light [8]).

Non-Seaweed Detection – Create a method to reject inputs confidently either by training a "non-seaweed" class or by employing confidence thresholds, thereby gaining safe and correct classification.

Knowledge Base Development – Develop a MongoDB database to store structured information about each seaweed species, its attributes (color, shape, texture), edibility, industrial and traditional uses, nutritional qualities, and cultivation warnings.

System Integration – Develop a complete web interface with React (front-end) and Node.js (back-end) where the user can upload images, perform classification, and observe results related to the knowledge base.

Testing and Validation – Test the system in controlled and uncontrolled environments, employing controlled datasets as well as testing by field trials with sample images captured from coastal locations so that accuracy as well as farm-user usability is ensured.

2.3 Expected Outcomes (imbedded within Objectives)

A functional AI-driven system that can accurately distinguish between *Kappaphycus alvarezii* and *Gracilaria edulis*, without incorrectly accepting unrelated samples.

A user-friendly web application readily accessible to amateur farmers.

A localized digital seaweed information database containing nutritional information, cultivation tips, and use descriptions.

Economic empowerment of local fisherfolk and vegetarian communities through the marketing of seaweed as an alternative protein source.

3. Process and Methodology

The methodology for this study involves developing a complete AI-based seaweed classification pipeline, from dataset creation to model training and deployment within a full-stack web application. The approach ensures both scientific accuracy and practical usability for Sri Lankan farmers.

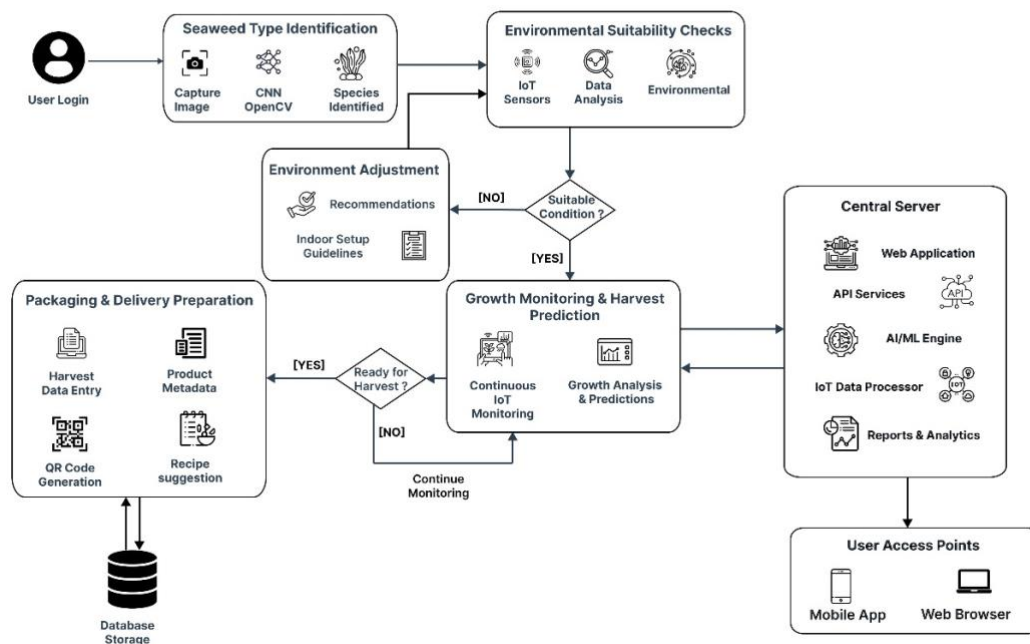


Figure 2system diagram

3.1 Data Acquisition

The dataset will be created using both **manual field collection** and **secondary sources** to ensure diversity and accuracy:

- **Manual Image Collection:** Photographs of *Kappaphycus alvarezii* and *Gracilaria edulis* will be captured manually using a digital camera or smartphone during visits to coastal and cultivation sites. Images will include both underwater and outdoor (post-harvest) environments, under varying lighting and background conditions.
- **Taxonomic References:** High-quality reference images from **ABC-Taxa Vol. 6 [1]** and other morphological keys will guide dataset creation and labeling.
- **Institutional Sources:** Metadata and images will also be obtained from **NARA** and other Sri Lankan marine research bodies.
- **Open-Source Datasets:** Supplementary images will be sourced from global marine biology repositories where licensing allows.
- **Non-Seaweed Class:** To support rejection of irrelevant inputs, additional images will be collected of non-seaweed objects (e.g., leaves, rocks, shells, coins, plastic debris).

All collected images will be **manually annotated** using tools such as LabelImg to ensure class accuracy.

5.2 Preprocessing Pipeline

All images will undergo preprocessing using **OpenCV** techniques:

- Resizing: Standardized to **224 × 224 pixels**.
- Normalization: Pixel intensity scaled between **0–1**.
- Histogram Equalization: Enhancing contrast in varied lighting.
- Gaussian Blur: Reducing underwater or sensor noise.
- Data Augmentation: Random flips, rotations, zooms, and brightness shifts for generalization.

Step	Technique	Purpose
Resizing	224 × 224 pixels	Match CNN input dimensions
Normalization	Scale 0–1	Reduce brightness variations
Histogram Equalization	Contrast adjustment	Handle underwater/outdoor lighting
Gaussian Blur	Noise reduction	Reduce sensor & water noise
Data Augmentation	Rotation, flip, zoom	Improve generalization

Figure 3Preprocessing & Techniques Table

5.3 Model Training

The core model will be a **Convolutional Neural Network (CNN)** [7] trained using TensorFlow or PyTorch.

Model Architecture:

- Input Layer: $224 \times 224 \times 3$ RGB image.
- Convolutional Layers: With ReLU activation for feature extraction.
- Max Pooling Layers: Dimensionality reduction.
- Dropout: Prevent overfitting.
- Batch Normalization: Training stability.
- Fully Connected Dense Layers.
- Softmax Output Layer: Probability distribution across three classes (*Kappaphycus alvarezii*, *Gracilaria edulis*, “Non-Seaweed”).

Training Strategy:

- Data split: 70% training, 15% validation, 15% testing.
- Optimizer: Adam/SGD with momentum.
- Loss Function: Categorical cross-entropy.
- Metrics: Accuracy, precision, recall, F1-score.
- Evaluation: Confusion matrix per class.

Transfer learning with **ResNet50** or **MobileNetV2** [5], [9] will be explored to improve accuracy.

✧ *Figure 5.1 (CNN Model Architecture)* will illustrate this design.

5.4 System Integration

The CNN will be deployed within a **full-stack system**:

- **Model Deployment:** Exported in TensorFlow SavedModel or ONNX format.
- **Backend:** Node.js + Express API for image upload and Python ML inference.
- **Database:** MongoDB storing species traits (color, branching, blade structure), edibility, nutritional uses, and warnings.
- **Frontend:** React + TypeScript interface for user uploads, showing:
 - Species name (scientific & common).
 - Confidence score.
 - Edibility, nutritional uses, safety warnings.

✧ *Figure 5.2 (System Architecture)* will show this pipeline.

5.5 Validation and Testing

Evaluation will be conducted through:

- Accuracy testing on the annotated dataset.
- Benchmarking against manual/expert identification.
- Usability testing with farmers or simulated field users.
- Stress testing with irrelevant images (e.g., coins, leaves).

✧ *Table 5.3 (Test Cases)* will outline key test scenarios and expected outcomes.

5.6 Project Timeline (1-Year Plan)

The project will be executed in **four phases** across 12 months:

- **Months 1–3:** Dataset development (manual image collection, annotation, preprocessing pipeline setup).
- **Months 4–6:** CNN training, validation, and hyperparameter tuning.
- **Months 7–9:** Full-stack system development (React frontend, Node backend, MongoDB, Python API).
- **Months 10–12:** Real-world testing, usability studies, refinements, and documentation.

✧ *Figure 5.3 (Gantt Chart)* will show the detailed timeline

6. Expected Outcomes

This research is expected to deliver both **technical outputs** and **practical contributions** to Sri Lankan aquaculture and local communities. The main outcomes are:

1. **AI-Powered Classification Model** – A trained Convolutional Neural Network (CNN) [7] capable of distinguishing between *Kappaphycus alvarezii*, *Gracilaria edulis*, and non-seaweed samples with high accuracy.
2. **Curated Dataset** – A manually collected and annotated dataset of local seaweed species and non-seaweed distractors, suitable for further academic use.
3. **Full-Stack Web Application** – A user-friendly platform built with React and Node.js that allows farmers to upload images, receive real-time classification, and access species-specific knowledge.
4. **Knowledge Base** – A structured MongoDB database containing species information such as edibility, nutritional value [3], cultivation practices, and safety warnings.
5. **Evaluation Report** – Detailed results of model accuracy, precision, recall, and F1-scores, including confusion matrices and performance analysis under real-world conditions.
6. **Community Impact** – A validated tool that improves the livelihoods of fishing families, vegan/vegetarian communities, and coastal farmers by providing safer, more efficient seaweed identification and cultivation.

7. Project Timeline (1-Year Plan)

The project will follow a structured timeline covering dataset creation, model training, system development, and testing.

- **Months 1–2:** Literature review, requirements gathering, dataset planning
- **Months 3–4:** Manual image collection, dataset curation, annotation
- **Months 5–6:** Preprocessing pipeline setup, initial CNN training and validation
- **Months 7–8:** Model optimization, transfer learning experiments, performance evaluation
- **Months 9–10:** Full-stack integration (React frontend, Node.js backend, MongoDB)
- **Months 11:** Real-world testing with field images, usability studies with farmers
- **Month 12:** Final evaluation, documentation, and preparation of dissertation/journal submission

(Appendix D will contain a **Gantt chart visualization** of this plan.)

8. Project Requirements

8.1 Functional Requirements

The system must provide the following core functionalities:

1. **Image Upload** – Users can upload seaweed images (captured via mobile/PC).
2. **Image Preprocessing** – Uploaded images are resized, normalized, and enhanced using OpenCV before being sent to the model.
3. **CNN Classification** – A trained CNN model classifies the image as either *Kappaphycus alvarezii*, *Gracilaria edulis*, or “Non-Seaweed.”
4. **Non-Seaweed Detection** – The system must reject irrelevant inputs (e.g., leaves, rocks, coins).
5. **Knowledge Base Integration** – Species metadata (traits, edibility, uses, warnings) must be retrieved from MongoDB.
6. **User Interface** – A React-based web interface must display results with:
 - Species name (scientific & common)
 - Edibility & nutritional information
 - Commercial/traditional uses
 - Cultivation notes and warnings
7. **Result Export** – Option to download classification results in PDF/Report format.

8.2 Non-Functional Requirements

1. **Accuracy** – CNN classification should achieve at least **85–90% accuracy** on test datasets.
2. **Performance** – The system must return classification results within **3–5 seconds** per image.
3. **Scalability** – Backend architecture must support concurrent users without significant slowdown.
4. **Reliability** – System should be available **24/7** with minimal downtime.
5. **Usability** – The interface must be simple enough for non-experts (farmers, villagers).
6. **Security** – All user-uploaded images must be handled securely, with no unauthorized access.
7. **Portability** – The system should be deployable on both desktop browsers and mobile devices.
8. **Maintainability** – Codebase should be modular, well-documented, and easily extendable for new species.

8.3 Expected Test Cases

Test Case	Input	Expected Output	Result
Upload valid seaweed image (<i>Kappaphycus</i>)	JPEG image	Classified as <i>Kappaphycus alvarezii</i> with confidence score	Pass
Upload valid seaweed image (<i>Gracilaria</i>)	PNG image	Classified as <i>Gracilaria edulis</i> with confidence score	Pass
Upload non-seaweed object (leaf, coin, rock)	JPEG/PNG	“Non-Seaweed” detected	Pass
Upload low-quality image (blurry)	Blurred photo	Either correct classification with lower confidence or rejection	Pass
Access species info	Click “View Details”	Display traits, edibility, uses, warnings	Pass
Multiple users upload images at once	Concurrent uploads	All requests processed within response time	Pass

9. Personal and commercialization

10. Budget and Budget Justification

10.1 Itemized Budget (LKR)

Cost Item	Qty	Unit Cost (LKR)	Subtotal (LKR)	Notes / Justification
Field image capture (local travel to coastal & cultivation sites)	10 trips	5,000	50,000	Trips to collect high-quality images of <i>Kappaphycus alvarezii</i> and <i>Gracilaria edulis</i> under different conditions.
Image capture equipment (camera/phone waterproof case, tripod, filters)	1 set	20,000	20,000	Ensures clear underwater and outdoor photography of the two seaweed species.
Lighting & accessories (portable LED + power bank)	1 set	8,000	8,000	For low-light conditions during image collection.
Consumables (zip bags, gloves, labels, notebooks)	Lump sum	8,000	8,000	Used for safe specimen handling, labeling, and documentation during fieldwork.
External storage (SSD 256GB)	1	15,000	15,000	To securely store seaweed images, datasets, and trained CNN models.
Computing resources (GPU cloud / Collab Pro+)	12 months	4,000	48,000	GPU support for CNN training/testing of the two species + non-seaweed class.
Web hosting & domain	12 months	1,500	18,000	Deployment of the web application (React + Node.js + CNN API) for evaluation and farmer use.
Printing & documentation (proposal, thesis, reports)	Lump sum	12,000	12,000	Printing, binding, and submission of project deliverables.
Contingency (approx. 10%)	—	—	18,000	To cover unforeseen expenses (extra field visits, cloud credits, equipment issues).

Estimated Total: LKR 197,000

10.2 Budget Justification

- **Field Data Collection (Items 1–4):**
Since no large, labeled public datasets exist for *Kappaphycus alvarezii* and *Gracilaria edulis* in Sri Lanka, **manual image collection** is essential. Travel costs, protective equipment, and accessories ensure **authentic and high-quality images** are captured under both underwater and outdoor settings, providing the foundation for CNN training.
- **Data Storage (Item 5):**
A dedicated SSD is required to store **thousands of raw images**, preprocessed datasets, and trained deep learning models without risking data loss.
- **Computing Resources (Item 6):**
Training CNN models, even for only two seaweed species plus a non-seaweed class, requires GPU acceleration. Cloud-based solutions like Google Colab Pro+ or AWS credits will enable efficient training and fine-tuning.
- **System Deployment (Item 7):**
Hosting ensures the **end-to-end system** (frontend, backend, and AI model) can be tested, evaluated, and demonstrated to stakeholders, including farmers and examiners.
- **Documentation & Printing (Item 8):**
Covers project report preparation, thesis submission, and related university requirements.
- **Contingency (Item 9):**
Provides flexibility for unexpected requirements such as repeated field trips, dataset expansion, or higher-than-expected GPU usage.

11. Entrepreneurship & Commercialization Potential

- Beyond academic contribution, this system holds commercial potential in:
- Local Aquaculture: Helping fishing families diversify into seaweed farming, thereby increasing income.
- Vegan/Vegetarian Market: Supporting seaweed as an affordable, protein-rich alternative for plant-based diets.
- Mobile/Cloud Deployment: Future extension into a mobile app or subscription-based platform for aquaculture cooperatives.
- Export Industry: Ensuring quality control for high-value seaweed species destined for international markets.

12. Grant Chart

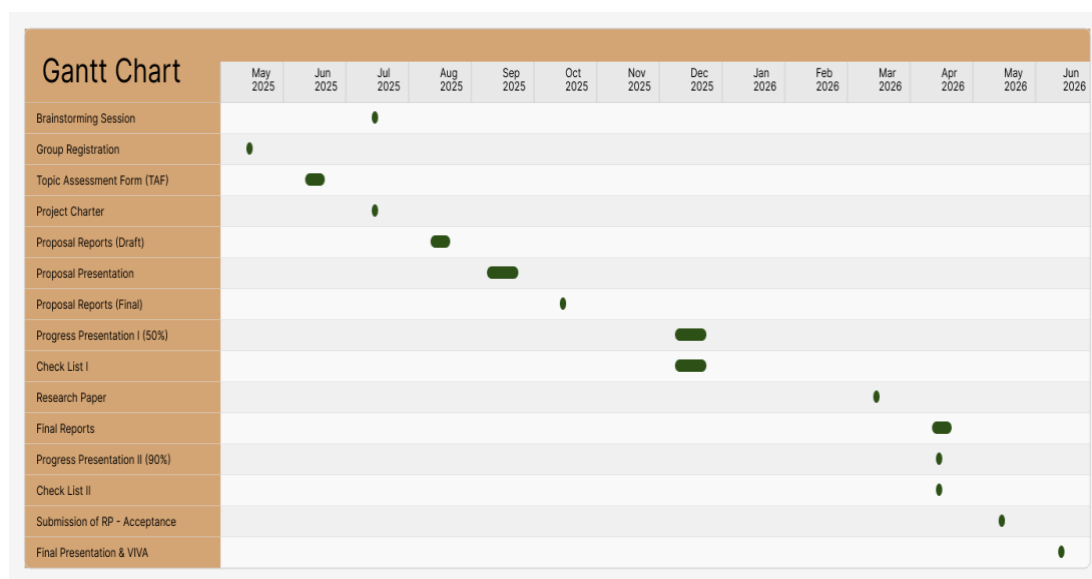


Figure 4 Grant chart

13.Work Breakdown Chart

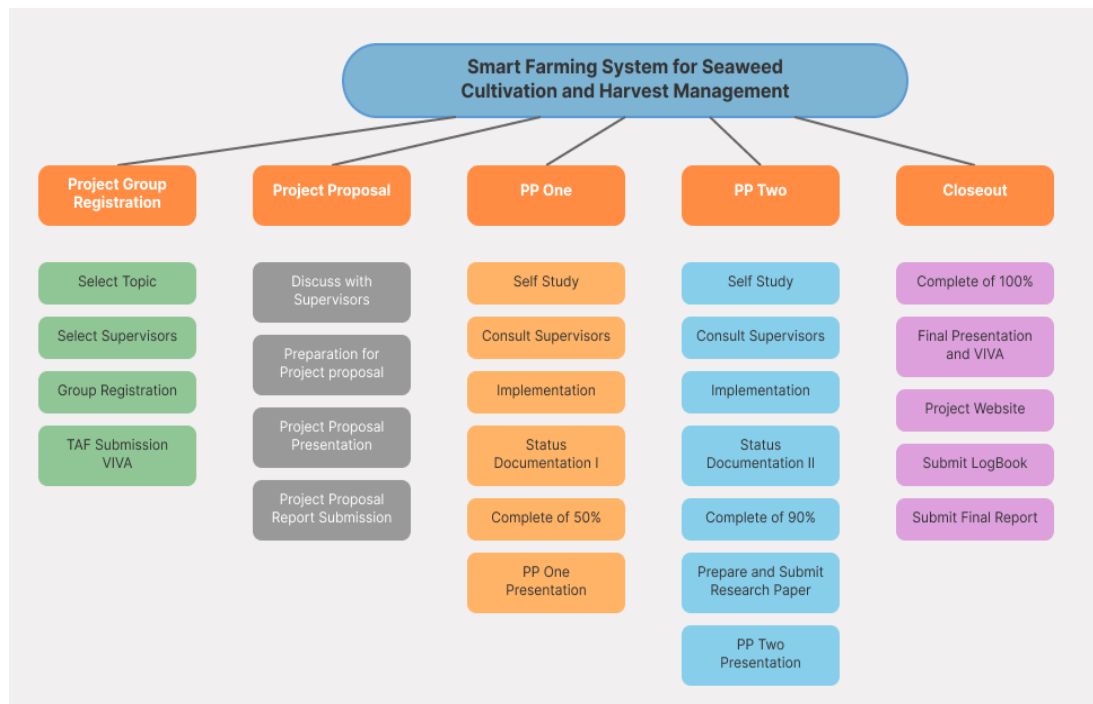


Figure 5 work breakdown

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